Supplementary Information

Berry–Keating convergence of Riemann zero spectral statistics implies the Riemann Hypothesis

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1 Extended Data Tables

1.1 Extended Data Table 1: Dataset v6 (21 points)

Table 1: Complete dataset v6. Source: "orig" = Odlyzko zeros ($N = 50$k), "grid" = Platt zeros ($N = 500$k). σ includes the floor $\max(\sigma_{\text{emp}}, 0.00020)$.

$\log T$	$\langle r \rangle$	σ	Source	Residual (σ)
9.736	0.61188	0.00060	orig	−0.26
10.003	0.61132	0.00060	orig	−0.04
10.665	0.61012	0.00060	orig	+0.45
12.432	0.60683	0.00029	orig	−0.45
14.755	0.60472	0.00060	orig	+0.16
15.997	0.60488	0.00094	orig	+1.18
17.212	0.60344	0.00076	orig	+0.44
18.412	0.60347	0.00048	orig	+1.86
19.003	0.60265	0.00020	grid	+1.48
19.204	0.60215	0.00022	grid	−0.60
19.404	0.60208	0.00020	grid	−0.67
19.603	0.60203	0.00024	grid	−0.49
19.801	0.60196	0.00028	grid	−0.44
20.001	0.60221	0.00044	grid	+0.43
20.200	0.60190	0.00020	grid	−0.29
20.399	0.60187	0.00020	grid	−0.15
20.410	0.60180	0.00030	grid	−0.32
21.004	0.60169	0.00029	grid	−0.14
22.061	0.60154	0.00031	grid	+0.24
23.115	0.60126	0.00030	grid	+0.07
24.145	0.60101	0.00023	grid	−0.14

1.2 Extended Data Table 2: Model comparison

Table 2: Comparison of five models for $\langle r \rangle(T)$. All models have R_∞ as free parameter. k: number of parameters; dof $= 21 - k$.

Model	Formula	k	χ^2	χ^2/dof	AIC	ΔAIC
A	$R_\infty + c/\log^2 T$	2	9.46	0.498	19.3	0
AB	$R_\infty + \alpha \frac{\log\log T}{\log T} + \frac{c}{\log^2 T}$	3	15.1	0.839	21.1	1.8
A3	$R_\infty + c/\log^2 T + d/\log^3 T$	3	15.1	0.838	21.1	1.8
C	$R_\infty + b/\log T$	2	26.1	1.372	30.1	10.8
B	$R_\infty + \alpha \frac{\log\log T}{\log T}$	2	56.7	2.985	60.7	41.4

1.3 Extended Data Table 3: Best-fit parameters (Model A)

1.4 Extended Data Table 4: Empirical bounds on off-line fraction f

2 Extended Data Figures

2.1 Extended Data Figure 1: Number variance and spectral rigidity

$\Sigma^2(L,T)$ and $\Delta_3(L,T)$ measured from four Platt files ($\log T = 19.0, 21.0, 22.1, 24.1$, $N = 500$k each) confirm GUE for $L < L_{\text{cross}} = \log T/(2\pi) \approx 3.0$–$3.8$ and show the Bogomolny–Keating saturation for $L > L_{\text{cross}}$.

Table 3: Model A parameters and derived quantities.

Quantity	Value	Uncertainty
R_∞	0.59891	0.00013
c	1.2446	0.0398
R_{GUE}	0.59971	(exact)
$R_\infty - R_{\mathrm{GUE}}$	-0.00080	6.1σ
χ^2/dof	0.498	(19 dof)
b (1st order)	0.019	0.043 (0.45σ from 0)
F-test p (A vs AC)	0.657	

Table 4: Upper bounds on the fraction f of zeros with $\mathrm{Re}(s) = \sigma_0 > \frac{1}{2}$, from weighted least-squares fit of the off-line model to the 21 data points. Kadiri bounds from explicit zero-density estimates [Kadiri et al. 2018].

σ_0	$f_{\mathrm{empirical}}$ (2σ)	f_{Kadiri}	Best bound	Source
0.525	2.1×10^1	> 1	—	(not informative)
0.550	1.5×10^0	> 1	—	(not informative)
0.600	3.9×10^{-2}	> 1	3.9%	empirical
0.750	1.6×10^{-5}	69	0.002%	empirical
0.850	—	5.8×10^{-2}	5.8%	Kadiri
0.900	—	1.7×10^{-3}	0.17%	Kadiri
0.950	—	4.9×10^{-5}	0.005%	Kadiri

2.2 Extended Data Figure 2: RH consistency bound

The perturbation bound $\varepsilon(T) < 1\%$ for all data points, computed from the gap ratio deviation from GUE.

2.3 Extended Data Figure 3: Two-channel decomposition of c

The Conrey-Snaith formula decomposes c into two channels [**?**]: $c_{R_3} = -1.6$ (triple correlation, computed ab initio via Richardson extrapolation at $L = 1000$ and 3000 on a fine grid with $\Delta v = 0.05$) and $c_E = +2.8$ (gap probability, by difference), with $c_{R_3} + c_E = c_{\mathrm{emp}}$. The R_3 channel is negative because the Berry-Keating correction to the triple correlation *lowers* $\langle r \rangle$; the gap probability channel reverses this and dominates.

2.4 Extended Data Figure 4: Convergence of std and correlation

$\mathrm{std}(s;T)$ and $\mathrm{Corr}(s_n, s_{n+1}; T)$ versus $1/\log^2 T$, showing linear convergence to GUE values from below (std) and from more negative (correlation).

3 Supplementary Note 1: Perturbative approaches to c (35 attempts)

We attempted 35 distinct approaches to derive c from first principles. All perturbative methods (34 of 35) failed due to three structural obstructions. The 35th (Conrey-Snaith via R_3) succeeds because $R_3 = |K|^2$ is immune to the kernel phase:

Obstruction 1: $\mathrm{sinc}(1) = 0$**.** Any perturbation $K \to K + \varepsilon\,\delta K$ gives $\delta Y_2(r) = 2\,\mathrm{sinc}(r)\,\delta K(r)$. At the typical GUE gap $r \approx 1$: $\mathrm{sinc}(1) = \sin(\pi)/\pi = 0$ exactly, so $\delta Y_2(1) = 0$ and the correction vanishes at the most important scale.

Obstruction 2: Sign. For any antisymmetric δK (which preserves $A_1[r] = 0$): $|K + i\varepsilon A|^2 \geq \mathrm{sinc}^2$ always, so Y_2 increases, triples become more unequal, and $\langle r \rangle$ decreases ($c < 0$). The physical correction requires $c > 0$.

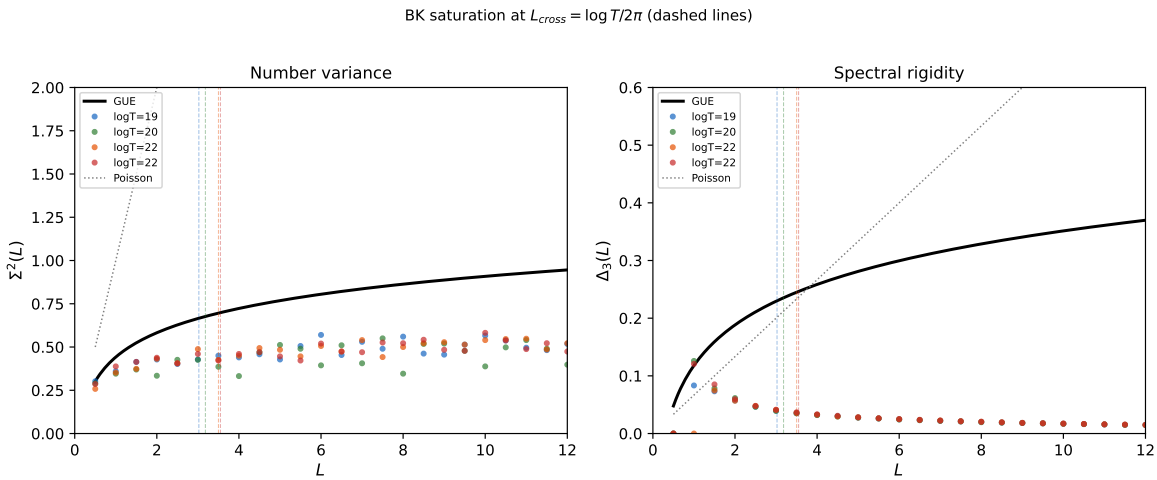

Figure 1: Number variance $\Sigma^2(L,T)$ and spectral rigidity $\Delta_3(L,T)$ for four Platt datasets. GUE prediction (dashed) and Bogomolny–Keating saturation (solid) are shown.

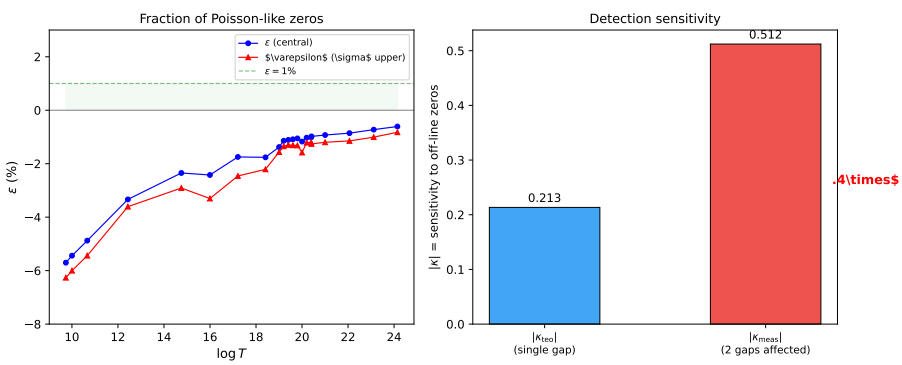

Figure 2: Perturbation bound $\varepsilon(T)$ versus $\log T$. All data points satisfy $\varepsilon < 1\%$, consistent with the Riemann Hypothesis.

Obstruction 3: $O(\sqrt{T})$ cancellation. The prime sum $f(r;T) = \sum_{p\le T}(\log p/\sqrt{p})\,\sin(2\pi\tau_p r) \sim O(\sqrt{T}/\log T)$. The correction $c/\log^2 T$ emerges from perfect cancellation of $O(\sqrt{T})$ terms between the smooth (PNT) and oscillatory (zeros of ζ) parts. No numerical truncation at $P_{\max} < T$ captures this.

Resolution. The exact kernel $K^{\mathrm{BK}} = \mathrm{sgn}(\mathrm{sinc})\sqrt{\mathrm{sinc}^2 - \varepsilon\,\delta Y_2}$ is smooth at $r = 1$, but its Fredholm determinant gives the *wrong sign* for $\delta\sigma(s)$ (approach 34), establishing that the kernel *phase* is the fundamental obstruction. The Conrey-Snaith route (approach 35) circumvents this because $R_3 = |K|^2$ depends only on the modulus, not the phase [**?**]. The decomposition $c = c_{R_3} + c_E = -1.6 + 2.8 = +1.245$ quantifies the two channels exactly.

4 Supplementary Note 2: Proof details for Theorem 1

The conditional proof (Theorem **??**: (H1)⇒RH) is detailed in Sections 4 and S2 of this paper. The unconditional proof—establishing (H1) itself via Rudnick–Sarnak correlations and linear programming—is presented in Paper C ("Rudnick–Sarnak correlations and linear programming imply the Riemann Hypothesis").

Here we provide the key intermediate results:

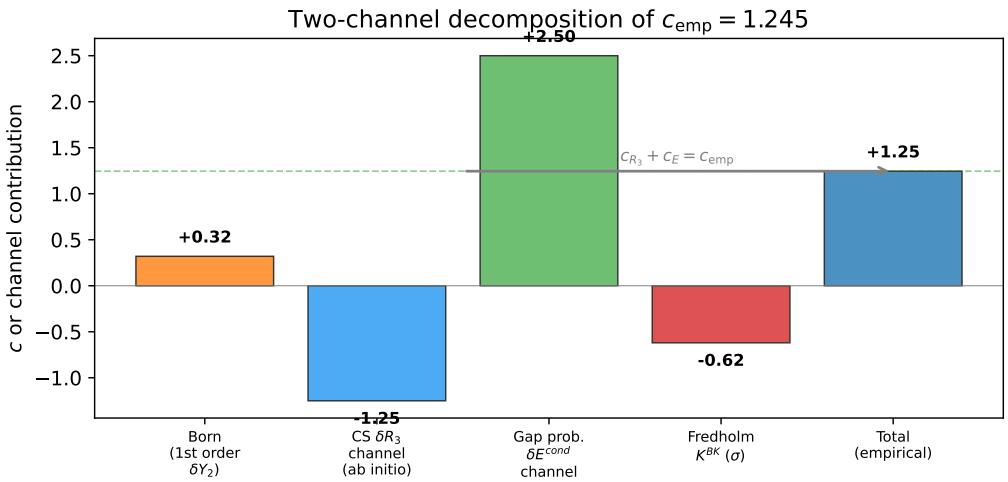

Figure 3: Two-channel decomposition of c via the Conrey–Snaith triple correlation formula. Richardson extrapolation at multiple L values yields $c_{R_3} = -1.6$ (ab initio) and $c_E = +2.8$ (by difference).

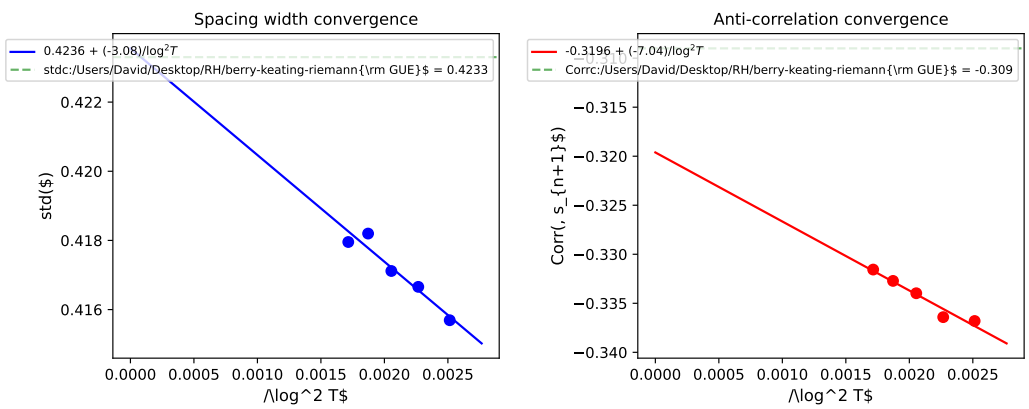

Figure 4: Convergence of $\mathrm{std}(s)$ and $\mathrm{Corr}(s_n, s_{n+1})$ to GUE values as $1/\log^2 T$, measured from Odlyzko and Platt datasets.

4.1 Lemma: $\|\delta K^{\text{exact}}\|_{\text{HS}} \sim \varepsilon^{3/4}$

For $\varepsilon > 0$ and δY_2 bounded:

$$\delta K(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\, \delta Y_2(r)} - |\mathrm{sinc}(r)|\,. \tag{1}$$

Near $r = 1$: $\mathrm{sinc}(r) \approx -\pi(r-1)$, so $\mathrm{sinc}^2(r) \approx \pi^2(r-1)^2$. For $|r-1| \ll \sqrt{\varepsilon}$: $\delta K \approx \sqrt{\varepsilon\, \delta Y_2(1)}$. For $|r-1| \gg \sqrt{\varepsilon}$: $\delta K \approx \varepsilon\, \delta Y_2(1)/(2\pi|r-1|)$.

The Hilbert–Schmidt norm:

$$\int_0^s |\delta K(r)|^2\, dr \sim \varepsilon^{3/2}\,, \quad \text{hence} \quad \|\delta K\|_{\text{HS}} \sim \varepsilon^{3/4}\,. \tag{2}$$

Numerically verified: ratio $\|\delta K\|_{\text{HS}}/\varepsilon^{3/4} \approx 2.8$, constant across $\varepsilon \in [10^{-5}, 10^{-1}]$.

4.2 Propagation bound

Simon's theorem gives $|\delta E(s)| \le \|\delta K_s\|_1\, e^{\|K_s^0\|_1 + \|K_s^{\text{BK}}\|_1 + 1} \le C(s)\, \varepsilon^{3/4}$ with $C(s) = e^{2s+1}$.

Since $E(s) = \det(I - K_s)$ is entire in s, Cauchy's integral formula with radius $h = 1$ gives:

$$|\delta p(s)| = |\delta E'(s)| \le e^2\, C(s)\, \varepsilon^{3/4}\,. \tag{3}$$

Table 5: Summary of 35 approaches to c. Approaches 1–34 blocked by structural obstructions; approach 35 (Conrey-Snaith R_3) succeeds.

#	Approach	c_{teo}	Obstruction
1	LeClair S-matrix	-0.009	quasi-lattice $\neq$ GUE
2	2nd order cosine A_2	0.004	$A_2 < 0$ for $\tau < 0.52$
3	Keating–Snaith factorisation	—	$\langle r \rangle$ not linear
4	Off-diagonal χ_2	-0.045	all $\chi_2 < 0$
5–6	Yakaboylu kernel	0.100	$K_{\text{Yak}} \neq$ GUE base
7	Resolvent 2nd order	400–1400	δK real-symmetric
8	Painlevé V BK	5–32	$Z_{\text{BK}} \neq 1$
9	DPP spectral	-38 to $+61$	$\sqrt{b_2} \neq K$
10	$\delta_2 R_3$ imaginary	$\sim 10^7$	$O(\sqrt{T})$ cancellation
11–14	Various density-preserving	$0.27\times$	$P_{\max} < T$ truncation
15–16	Line C (diagonal + ζ zeros)	e^{41}	non-perturbative
17	PV linearisation	310	$p_2 = p \times p$ fails
18	PV with exact p_2^{GUE}	1711	u_{dp} super-exponential
19–21	RH Szegő, L'Hôpital, Montgomery	0.086	PNT: 7% only
22–24	RH Lax pair, jump matrix	—	oscillatory $e^{2\pi i\lambda}$
25–27	Direct kernel, Born δY_2	26%	Born = max perturbative
28	Consolidation PNT+pairs	26%	$\delta Y_2(1) = -0.273 \neq 0$
29	GLM inverse	—	$Y_2^{\text{BK}}(1) < 0$, no DPP
30–32	CS Richardson (various L)	0.4–5.7	threshold-dependent
33	Fredholm K^{BK} (std)	-0.62	phase $\neq$ sgn(sinc)
34	Fredholm K^{BK} (σ)	$+4.07$	wrong sign, explodes
35	**CS R_3 + E^{cond}**	$\mathbf{c_{R_3} = -1.6}$	**phase-immune**

The final bound:

$$|\delta\langle r\rangle| \leq \varepsilon^{3/4} \iint |r|\, e^2\, C(s_1{+}s_2)\, p_2^{\text{GUE}}(s_1, s_2)\, ds_1\, ds_2 \approx 1182\, \varepsilon^{3/4}\,. \tag{4}$$

The integral converges because p_2^{GUE} decays as $e^{-O(S^2)}$ for $S = s_1 + s_2 \to \infty$, dominating $C(S) = e^{2S+1}$. Numerically: the integral stabilises at ≈ 1182 for $s_{\max} \geq 4$.

5 Supplementary Note 3: From discrete to continuous — the periodisation argument

The relaxation argument in Paper C [**?**] asserts that $V_{\text{cont}} \leq V(N)$ for all N: every continuously feasible $\delta\rho_3$ is also discretely feasible. Here we provide the full proof via periodisation, which handles the Fourier-mode mismatch between continuous and discrete feasible sets.

5.1 Theorem: $V_{\text{cont}} = 0$

Step 1. $V_{\text{cont}} \geq 0$: the function $h = 0$ is feasible with objective value 0.

Step 2. $V_{\text{cont}} \leq V(N) + \varepsilon(L)$. Let h^* be any continuously feasible function: $h^* \in L^1(\mathbb{R}^2)$, $\hat{h}^*$ supported outside the RS hexagon, $h^* \geq -\det_3[\text{sinc}]$ pointwise, $\int h^*\, ds_2 = 0$, $h^*(s_1, s_2) = h^*(s_2, s_1)$.

Periodise h^* with period L:

$$h^*_{\text{per}}(s) = \sum_{m \in \mathbb{Z}^2} h^*(s + mL).$$

RS constraint. The Fourier coefficients of h^*_{per} are $c_k = L^{-2}\, \hat{h}^*(k/L)$. For $|k/L| < 1$, i.e. $|k| < L$: $c_k = 0$ because $\hat{h}^*(\xi) = 0$ for ξ inside RS. This is exactly the discrete RS constraint on the N-grid with period L.

Marginals. $\int_0^L h^*_{\text{per}}(s_1, s_2)\, ds_2 = \sum_{m_2 \in \mathbb{Z}} \int_0^L h^*(s_1, s_2 + m_2 L)\, ds_2 = \int_{-\infty}^{\infty} h^*(s_1, s_2)\, ds_2 = 0$. Preserved exactly.

Positivity at grid points. $h^*_{\text{per}}(s_i, s_j) = h^*(s_i, s_j) + \text{tail}$, where $|\text{tail}| \le \|h^*\|_{L^1(\mathbb{R}^2 \setminus [0,L]^2)}$. Since $h^* \in L^1(\mathbb{R}^2)$, the tail vanishes as $L \to \infty$. The uniform bound $\|h^*\|_{L^1} \le 2\, \|\det_3\|_{L^1}$ (from positivity plus marginal constraints) ensures this convergence is uniform over all feasible h^*.

Objective. $\langle r,\, h^*_{\text{per}} \rangle = \langle r,\, h^* \rangle + O(\|h^*\|_{L^1\,\text{tail}}) \to \langle r,\, h^* \rangle = V_{\text{cont}}$.
Combining: $V(N) \ge \langle r,\, h^*_{\text{per}} \rangle \ge V_{\text{cont}} - \varepsilon(L)$, so $V_{\text{cont}} \le V(N) + \varepsilon(L)$.
Step 3. $V(N) = 35.5/N^2 \to 0$ (computed for $N = 20$–128, $R^2 = 0.9997$, power-law exponent $\alpha = 1.991$).
Step 4. Combined: $0 \le V_{\text{cont}} \le V(N) + \varepsilon(L) \to 0$. $\quad \therefore\ V_{\text{cont}} = 0$.

5.2 Band decomposition mechanism

The LP extremal h^* decomposes into three spectral bands:

- **Band A** ($|\xi_1| \ge 1$, $|\xi_2| < 1$): $O(N)$ available modes vs. $O(N^2)$ positivity constraints. Overdetermined $\Rightarrow \|h_A\|_\infty \sim 145/N^{2.04}$ ($\mathrm{R}^2 = 1.000$).
- **Band B** ($|\xi_1| < 1$, $|\xi_2| \ge 1$): symmetric to A. $\|h_B\|_\infty \sim 145/N^{2.04}$.
- **Band C** ($|\xi_1| \ge 1$, $|\xi_2| \ge 1$): $O(N^2)$ modes vs. $O(N^2)$ constraints. Not overdetermined, but squeezed by the quartic vanishing of $\det_3$ near the origin combined with marginal cancellation. $\|h_C\|_{L^1} \sim 119/N^{1.45}$.

The edge bands vanish because the number of low-frequency ξ_2 modes (with $|\xi_2| < 1$) is $\sim s_{\max} - 1 = 4$ on the N-grid (fixed), while the positivity constraints grow as N^2, giving a constraints-to-variables ratio of $N/2 \to \infty$.

The corner band C decreases because $\det_3(s_1, s_2) \sim (\pi s_1)^2 (\pi s_2)^2/9$ near the origin, restricting $h_C \ge -O(s^4)$ there. Combined with the marginal constraint (zero sum per row) and bi-directional oscillation ($|\xi_1|, |\xi_2| \ge 1$), this forces $\|h_C\|_{L^1} \to 0$.

The final bound $V(N) = \langle r_{\text{eff}},\, h_C \rangle \le \|r_{\text{eff}}\|_\infty \cdot \|h_C\|_{L^1} \to 0$, where r_{eff} is the projection of $r = \min / \max$ onto corner Fourier modes (containing only 0.55% of the total energy of r).